

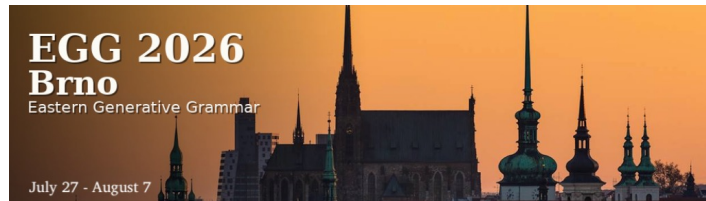


The Functional Architecture of Nominals

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<https://kxhoge.github.io/resources.html>



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Course outline

Monday	Approaches to the functional structure of nominals
Tuesday	The NP vs. DP debate
Wednesday	The NP vs. DP debate (cont.)
Thursday	Pronouns
Friday	D(emonstrative)-Pronouns

D(emonstrative)-Pronouns

Outline

1. D-pronouns vs. personal pronouns
2. Experimental evidence for a uniform approach to German d- and p-pronouns
3. Pronouns, demonstratives, and pronouns as demonstratives
4. Summary and taking stock

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- An early implementation of the idea that person and number features are associated with distinct functional heads is found in Wiltschko (1998):

‘Let us assume that AgrD [Wiltschko’s (1998) name for the category hosting phi-features] can be split into NumP and PersonP, as shown below:

... [DP [PersP [NumP [NP]]]]’
(Wiltschko 1998: 155).

- (1) a. ihnen [German]
3.pl.dat
‘them’
- b. [_{PersP} ihn [_{NumP} -en]]
(Wiltschko 1998: 155, ex. 22c))

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- Wiltschko (1998) investigates the syntactic properties of personal pronouns and so-called d-pronouns.
- A **d-pronoun** is a type of demonstrative pronoun whose form is morphologically identical to the corresponding definite article.

(2) a. Maria hat ihn / **den** gesehen.
Mary has him / d-pron seen
'Mary has seen him.'
(Wiltschko 1998: 144, ex. (1))

b. Maria hat **den** Mann gesehen.
Mary has the man seen
'Mary has seen the man.'

(3) Sie → / die → kann das besser.
she / d-pron can that better
'She is better at it.'

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- Although personal pronouns and d-pronouns overlap in certain contexts, they do not have an identical distribution:

- (4) a. Peter_i hat geglaubt, daß **er**_i/ ***der**_i dumm ist.
Peter_i has believed that he_i/ d-pron_i stupid is.
Peter_i has believed that he_i is stupid.
- b. Jeder Mann_i glaubt, daß **er**_i/ ***der**_i dumm ist.
Every man_i believes that he_i/ d-pron_i stupid is.
Every man_i believes that he_i is stupid.
- c. Peter bewundert den Mann, **der**/ ***er** das Bier erfand.
Peter admires the man d-pron/ he the beer invented.
Peter admires the man who invented beer.

(Wiltschko 1998: 144, ex. (2), (3) & (5))

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- For some speakers of German (including Wiltschko), personal pronouns may agree either grammatically or semantically with their antecedent, whereas d-pronouns can agree only grammatically.

- (5) a. Ein Mädchen_i kam zur Tür herein. **Sie_i**/ **es_i** war schön.
a girl_i came to-the door herein. She_{i,f}/ it_{i,n} was beautiful.
A girl came through the door. She was beautiful.

(Wiltschko 1998: 163, ex. (33))

- b. Ein Mädchen_i kam zur Tür herein. **Das_i**/ ***die_i** war schön.
A girl_i came to-the door here-in. The_{i,n}/ the_{i,f} was beautiful.
A girl came through the door. She was beautiful.

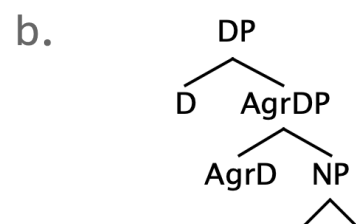
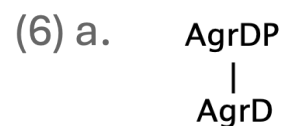
(Wiltschko 1998: 164, ex. (34))

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- Wiltschko (1998) derives the distributional differences between personal pronouns and d-pronouns from the fact that the two have different structural representations:

‘the former are merely the spell-out of phi-features ... whereas the latter are instances of the definite determiner containing an elliptical (empty) NP’ (Wiltschko 1998: 179).



D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- Accordingly, d-pronouns, but not personal pronouns, share a structure with full DPs.

(7) a. [DP d- [AgrD er [NP Mann / Ø]
 th- 3.M.SG.NOM man.M.SG
 ‘the man’ / d-pron.3.M.SG

 b. [AgrD *er*]
 3.M.SG
 ‘he’
 (cf. Wiltschko 1998: 149, ex. (10))

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- Additional empirical support for the analysis of personal pronouns as ϕ Ps and d-pronouns as structurally richer DPs comes from the morphological paradigms of German.

(8)

	Definite	Determiners	Personal	Pronouns
sg.	m.	n.	m.	n.
NOM	d-er	d-as	er	es
DAT	d-em	d-em	ihm	ihm
ACC	d-en	d-as	ihn	es

(Wiltschko 1998: 149)

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- If d-pronouns are full DPs, it follows that
 - they cannot be bound without giving rise to a Binding Theory Condition C violation;

- (9) a. Peter_i hat geglaubt, daß **er_i**/ ***der_i** dumm ist.
Peter_i has believed that he_i/ d-pron_i stupid is.
Peter_i has believed that he_i is stupid.
b. *Peter_i believed that [the man]_i was stupid.

- they cannot function as a bound variable;

- (10) a. Jeder Mann_i glaubt, daß **er_i**/ ***der_i** dumm ist.
Every man_i believes that he_i/ d-pron_i stupid is.
Every man_i believes that he_i is stupid.
b. *[Every man]_i believes that [the man]_i is stupid.

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- they can function as relative pronouns because the empty NP can be identified with the head noun of the relative clause;

(11) a. Peter bewundert den Mann, **der**/ ***er** das Bier erfand.
Peter admires the man d-pron/ he the beer invented.
Peter admires the man who invented beer.

b. Peter bewundert den [Mann]_i, [DP d- [AgrDP er [NP Mann]_i]] das Bier erfand.

D(emonstrative)-Pronouns

D-pronouns vs. personal pronouns

- they must exhibit grammatical rather than semantic agreement as the empty NP inherits the lexical entity and grammatical gender of its discourse antecedent.

(12) a. Ein Mädchen_i kam zur Tür herein. **Das_i/ *die_i** war schön.
A girl_i came to-the door here-in. The_{i,n}/ the_{i,f} was beautiful.
A girl came through the door. She was beautiful.

b. Ein [Mädchen]_N ... [DP d- [AgrDP as [NP Mädchen]_N]] war schön.

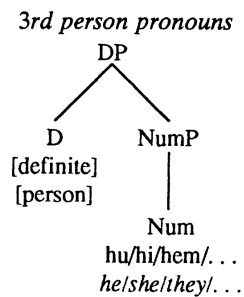
c. Ein [Mädchen]_N ... *[DP d- [AgrDP ie [NP Mädchen]_N]] war schön.

D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns

- Various analyses of the structure of third-person pronouns have been proposed in the literature.

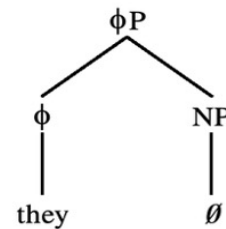
(13) a. Ritter (1995)



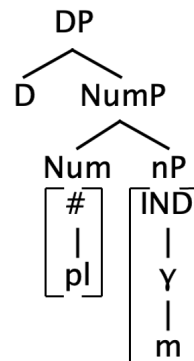
b. Wiltschko (1998)



c. Déchaine & Wiltschko (2002)

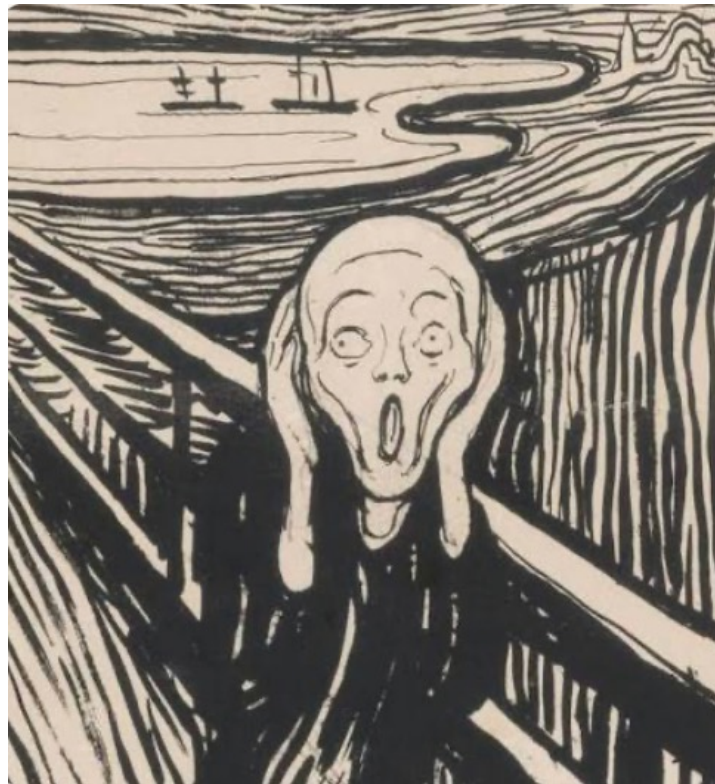


d. cf. Puškar-Gallien (2019)



D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns



D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns

- Can additional evidence help adjudicate between these structures?
- Patel-Grosz & Grosz (2017) argue that experimental findings concerning pronominal interaction with the Overt NP Constraint suggest that personal pronouns and d-pronouns do not differ in the availability of a null NP.

D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns

- Overt NP Constraint:
Donkey-anaphora ‘exhibit a preference for an explicit NP antecedent, which must not be part of another word’ (Patel-Grosz & Grosz (2017: 266); cf. Postal 1969).

(14) a. Every owner of a donkey_i beats it_i.

b. ?(*)Every donkey_i-owner beats it_i.

(Heim 1982: 80)

D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns

- The interpretation of donkey-anaphora involves a descriptive content to be recovered from the antecedent.
- Descriptive content can be recovered structurally or pragmatically (Hankamer & Sag 1976).

(15) Adam went to Brno and Zoe [did too].

- *do too* is a surface anaphor;
the anaphor is structurally resolved (by recovering the VP *go to Brno*).

(16) Adam went to Brno and Zoe [did that].

- *do that* is a deep anaphor;
the anaphor is pragmatically resolved.

D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns

- German allows both personal and d-pronouns to function as donkey-anaphora.

(17) Jeder Besitzer eines Esels_i schlägt ihn_i / den_i. [German]
every owner of-a donkey hits him / d-pron.
'Every owner of a donkey hits it.'

- 'If DEMs are [+ NP] and PERs [- NP], the prediction is that the former are structurally resolved ... and the latter pragmatically, in the spirit of Hankamer and Sag's (1976) and Sag and Hankamer's (1984) surface anaphora vs. deep anaphora. This would predict that the Overt NP Constraint should be stronger in the structurally resolved DEMs than in the pragmatically resolved PERs. By contrast, if such a distinction is missing, we expect the two types of pronouns to exhibit parallel effects of the Overt NP Constraint' (Patel-Grosz & Grosz 2017: 266).

D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns

- (18) Wenn eine Studentin {**Führerschein**besitzerin / Besitzerin eines **Führerscheins**} ist,
if a student driver's.license.owner owner of.a driver's.license is
dann trägt sie {**ihn** / **den**} meist im Geldbeutel mit sich.
then carries she PER DEM mostly in.the wallet with self
'If a student is {a **driver's license** owner / an owner of a **driver's license**}, then she
usually carries {**it** / **DEM**} around in her wallet.'
- The non-uniform analysis of personal pronouns and d-pronouns predicts a statistical interaction of pronoun and overtiness of antecedent: the effect of overtiness (i.e. the preference for *Besitzerin eines Führerscheins* 'owner of a driver's license') should be stronger with the d-pronoun than with the personal pronoun.

D(emonstrative)-Pronouns

Experimental evidence for a uniform approach to German d- and p-pronouns

- But the results show there to be **no statistical interaction** of pronoun type and overtness of antecedent, i.e. the effect of the Overt NP Constraint is no stronger with d-pronouns than it is with personal pronouns.

(19)

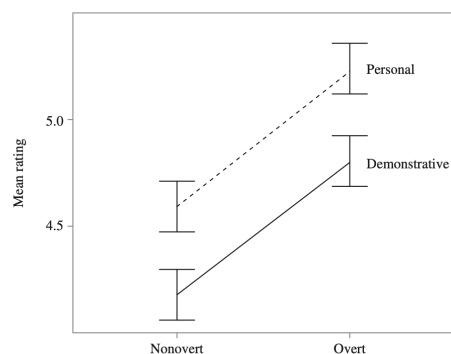


Figure 1
Mean ratings by condition

(Patel-Grosz & Grosz 2017: 268)

- Patel-Grosz & Grosz (2017) conclude that both types of German pronouns have the same core structure.

D(emonstrative)-Pronouns

Pronouns, demonstratives, and pronouns as demonstratives

- Pronouns and demonstratives could instantiate the same syntactic configuration while differing in the denotation of D.

D(emonstrative)-Pronouns

Pronouns, demonstratives, and pronouns as demonstratives

- In Ahn (2022), personal pronouns and definite descriptions are analysed as realising a semantic structure containing a supremum operator, while demonstratives are proposed to realise a semantic structure with a bi-supremum operator.
- A supremum operator maps a property P to the largest individual satisfying P .
Suppose a pronoun denotes a property of being identical to the [feminine] individual assigned to index 1:
$$\lambda x.x \ \& \ \text{feminine}(x) = g(1).$$
Assume that the extension of this property is the singleton set $\{\text{Mary}\}$.
Applying the supremum operator then yields the unique individual that satisfies the property:
$$\text{sup}(\{\text{Mary}\}) = \text{Mary}.$$

D(emonstrative)-Pronouns

Pronouns, demonstratives, and pronouns as demonstratives

- A bi-supremum operator maps a property P to the largest individual satisfying P and an additional restriction R.

Suppose a demonstrative denotes a property of being a [feminine] individual:

$$\lambda x.x \ \& \ [\text{feminine}](x).$$

Assume that the extension of this property is the set {Mary, Norma, Olga}.

The demonstrative component contributes an additional contextual restriction, corresponding to the speaker's pointing gesture.

Suppose that the speaker is pointing to Olga.

This restricts the extension from {Mary, Norma, Olga} to {Olga}.

The bi-supremum operator then selects the maximal individual satisfying both restrictions:

$$\text{bisup}(\{\text{feminine}(x)\}, \{\text{pointed-at individuals}\})$$

which yields

$$\text{sup}(\{\text{Olga}\}) = \text{Olga}.$$

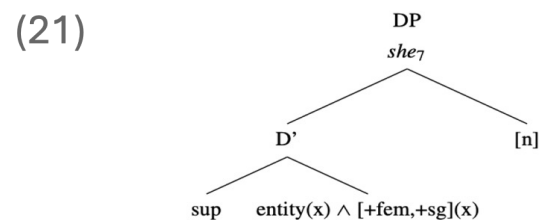
D(emonstrative)-Pronouns

Pronouns, demonstratives, and pronouns as demonstratives

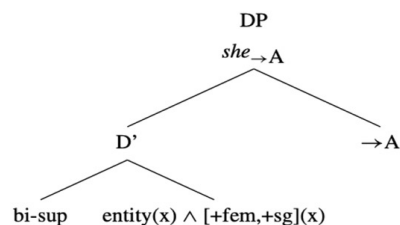
- Ahn (2022), assumes that personal pronouns can instantiate the demonstrative structure when used deictically (and they are not blocked by a competing demonstrative).

(20) a. (I saw Jane yesterday.)
She is moving to Brazil.

b. (Pointing to Jane)
She_→ is moving to Brazil.



(Ahn 2022: 1380, ex. (88) & (89))



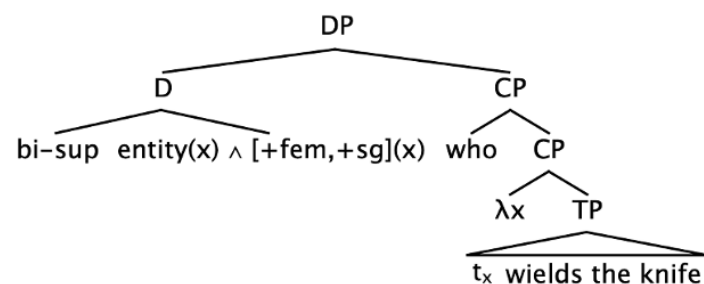
(22) (Pointing to Jane)
#That_→ is moving to Brazil.

D(emonstrative)-Pronouns

Pronouns, demonstratives, and pronouns as demonstratives

- Ahn (2022) further proposes that the second argument of the demonstrative bi-supremum operator can not just be a deictic pointing, but also an anaphoric index, or a relative clause.

(23) a



(Hoge, to appear, ex. (28))

b. she who wields the knife (never wears the crown)

D(emonstrative)-Pronouns

Pronouns, demonstratives, and pronouns as demonstratives

- If a personal pronoun can be used in place of a demonstrative only when no demonstrative with the relevant featural specification is available, then the asymmetrical behaviour of personal pronouns in pronoun–noun constructions derives from the availability of the bi-supremum structure, rather than from any structural differences between pronouns of different persons.

- (24) a. we (who are) linguists
b. you (who are) linguists
c. *they (who are) linguists
d. those linguists

- (25) a. ?them (who are) linguists

- (26) a. (Pointing to groups of people) I like them_→ but not them_→.
b. (Pointing to groups of people) #I like those_→ but not those_→.
(Ahn 2022: 1381, ex. (91))

D(emonstrative)-Pronouns

Summary and taking stock

- The move to ascribe different syntactic structures to pronouns of different types has led to a proliferation of phrase-structural representations for personal pronouns and d(emonstrative)-pronouns, despite a lack of experimental evidence corroborating the postulated structural differences.
- An alternative proposal treats pronouns as structurally identical, differing only in the abstract functional head realised in D.

D(emonstrative)-Pronouns

Summary and taking stock

- *What will I have learnt by the end of this week?*

Upon completion of the course, you will

- be familiar with the historical trajectory of the expansion of functional structure within nominal projections;
- have gained an understanding of the theoretical debate on the phrase structure of nominals;
- be able to evaluate arguments for and against postulating functional projections in the nominal domain;
- have acquired the theoretical background that is necessary to understand and situate current research on nominal syntax.

D(emonstrative)-Pronouns

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